

Partial report. Selected-run evidence status: stopped_early

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

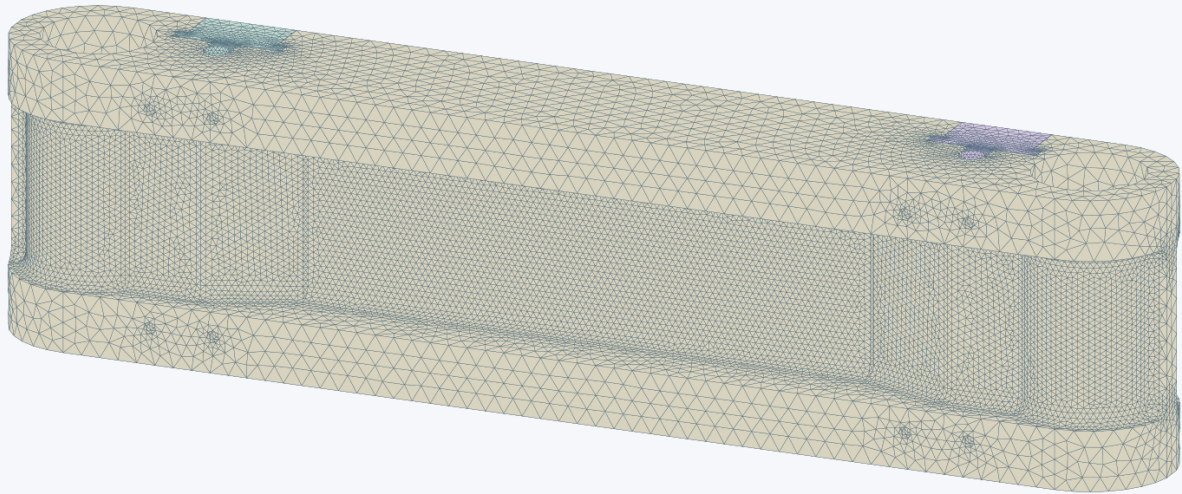
Mesh

Mesh overview and local detail

Mesh nodes	71447	
Tetrahedral elements	299858	
Meshed parts	3	
Component target sizes	5 / 3	mm
Minimum mean-ratio quality	0.309	0-1

Mesh overview

Undeformed exterior mesh

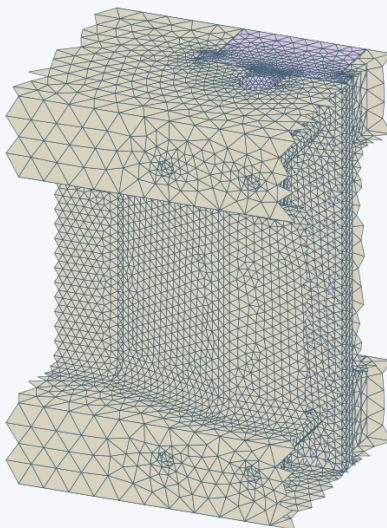


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Local mesh detail

Local mesh detail



Exterior mesh faces • undeformed geometry

Local mesh detail. Local view of the selected mesh; no local refinement claim is made.

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.05	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Termination time-step ratio	0.01	dimensionless
Printed parts mass	0.5627641	kg
Hardware mass	0.03400217	kg
Motor mass	0.16	kg
Moving mass	0.7567663	kg
Initial kinetic energy	945.9578	J

Recorded material, contact and boundary assumptions

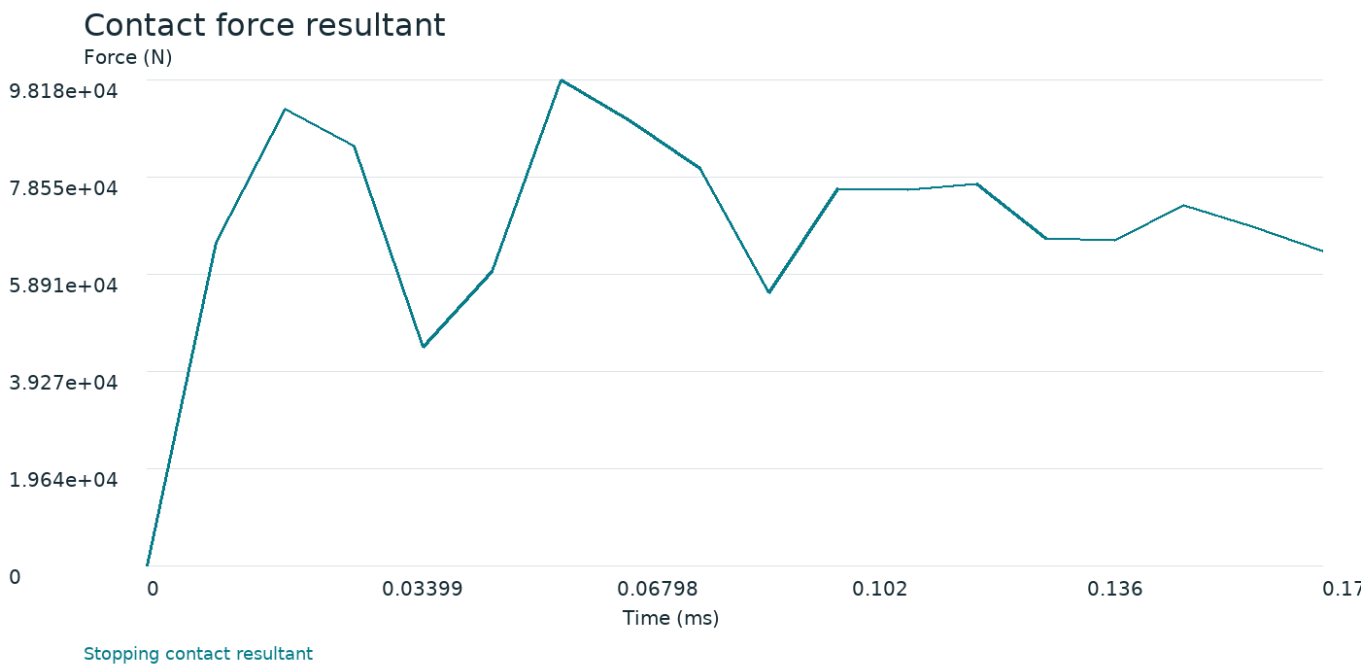
- Unchanged prior clamshell fixture and synthetic 54.46 kg/m³ MAT057 foam; not the proposed 30 kg/m³ product.
- Three independent printed solids; body-to-closure frictional contact plus eight ideal bolt collars; no slip/preload/failure capacity.
- Eight CAD generic steel bolt/nut envelope masses distributed on the collar nodes.
- Two 80 g motor masses distributed on the original motor bores; no casing stiffness.
- PA12-CF10 isotropic linear elastic screen; not printed anisotropic strength qualification.
- Ideal transverse cable-bore guidance leaves axial Z free; no cable compliance, tension or clearance contact.
- Two stationary aluminum spreaders, 0.1 mm initial gap; fixed foam rear Z, free foam sides.

- No carrier, motor thrust, element failure, fracture or damage.

Solve

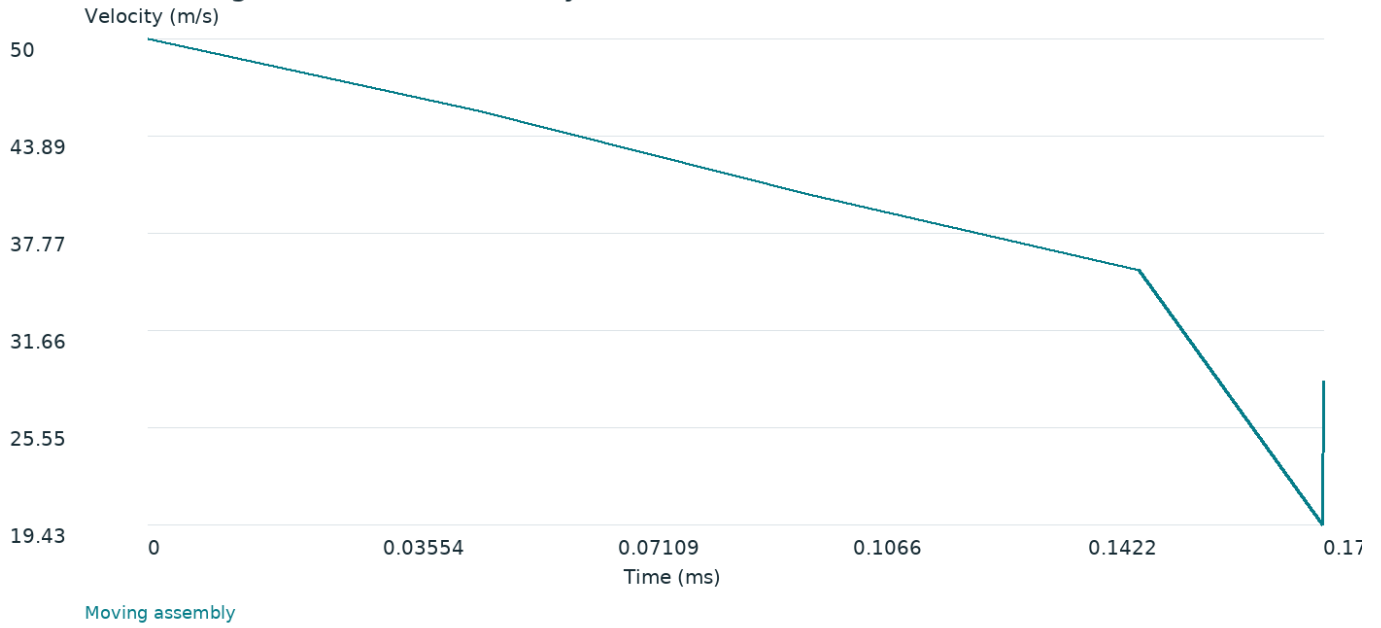
Recorded run status	stopped_early	
Recorded reached simulation time	0.169952	ms
Recorded requested simulation time	8	ms
Recorded solver wall time	226.094	s
Recorded termination state	normal_termination_confirmed	

Separate solve charts



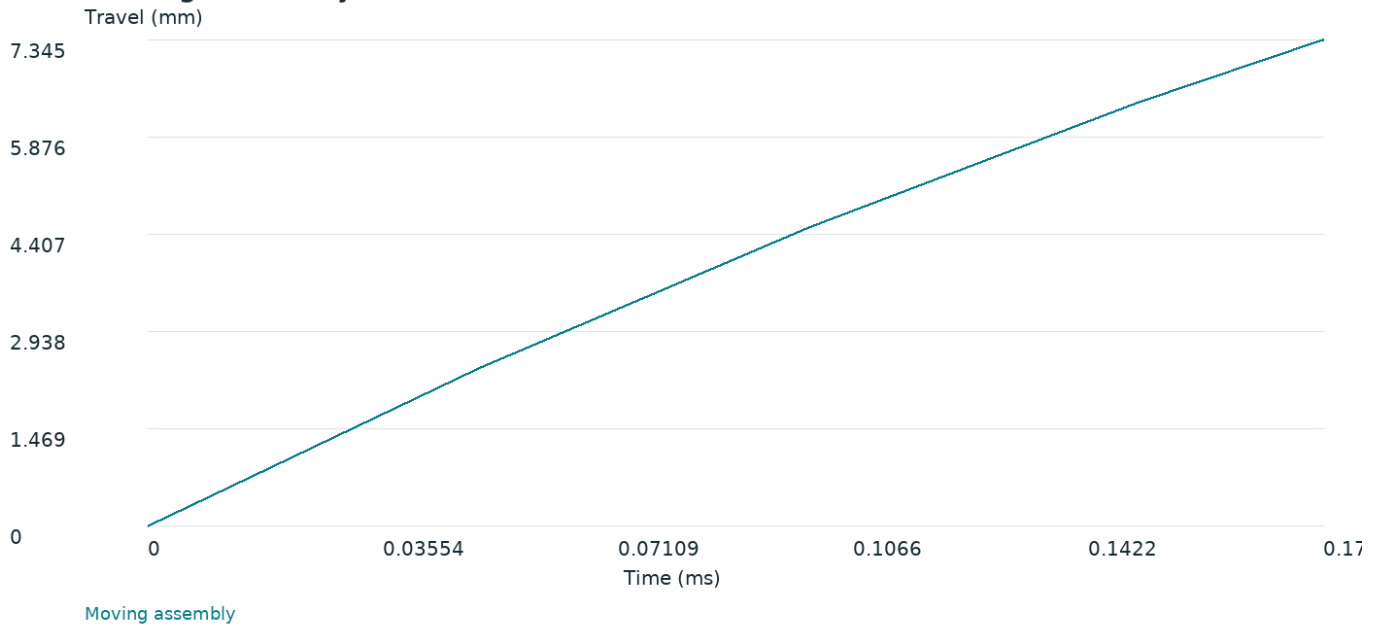
Contact force resultant. [projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/contact_histories.csv](#)

Mass-weighted forward velocity



Mass-weighted forward velocity.
projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/time_histories.csv

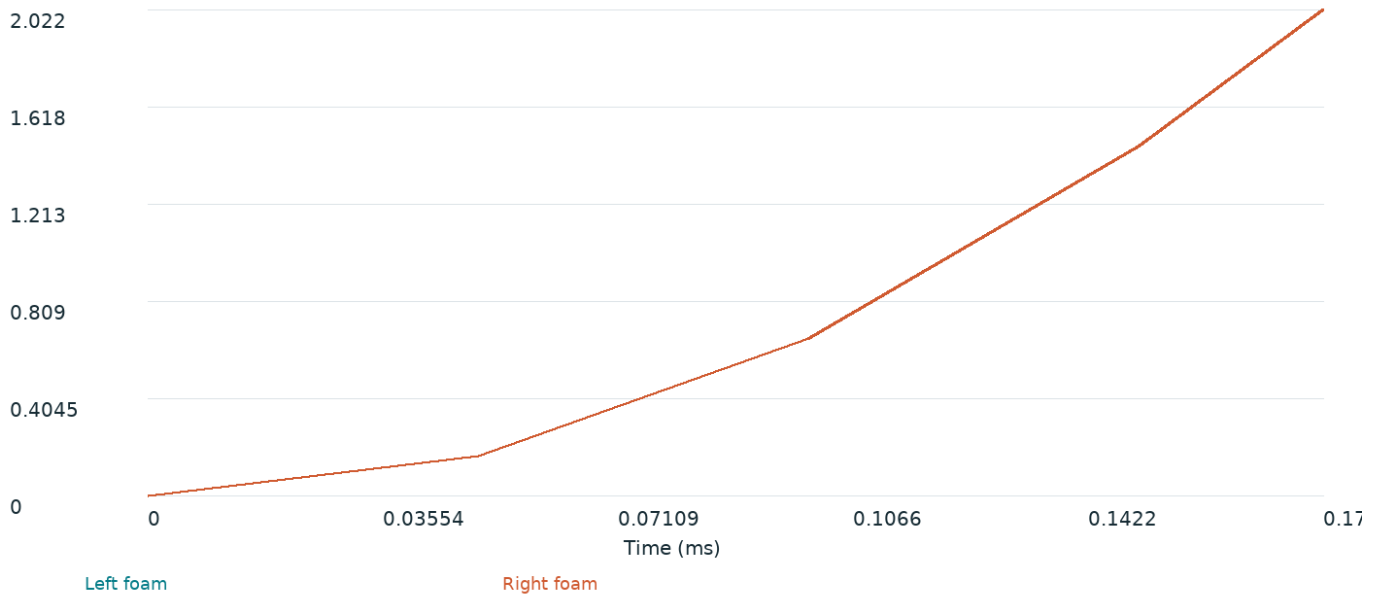
Moving assembly travel



Moving assembly travel. projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/time_histories.csv

Foam compression

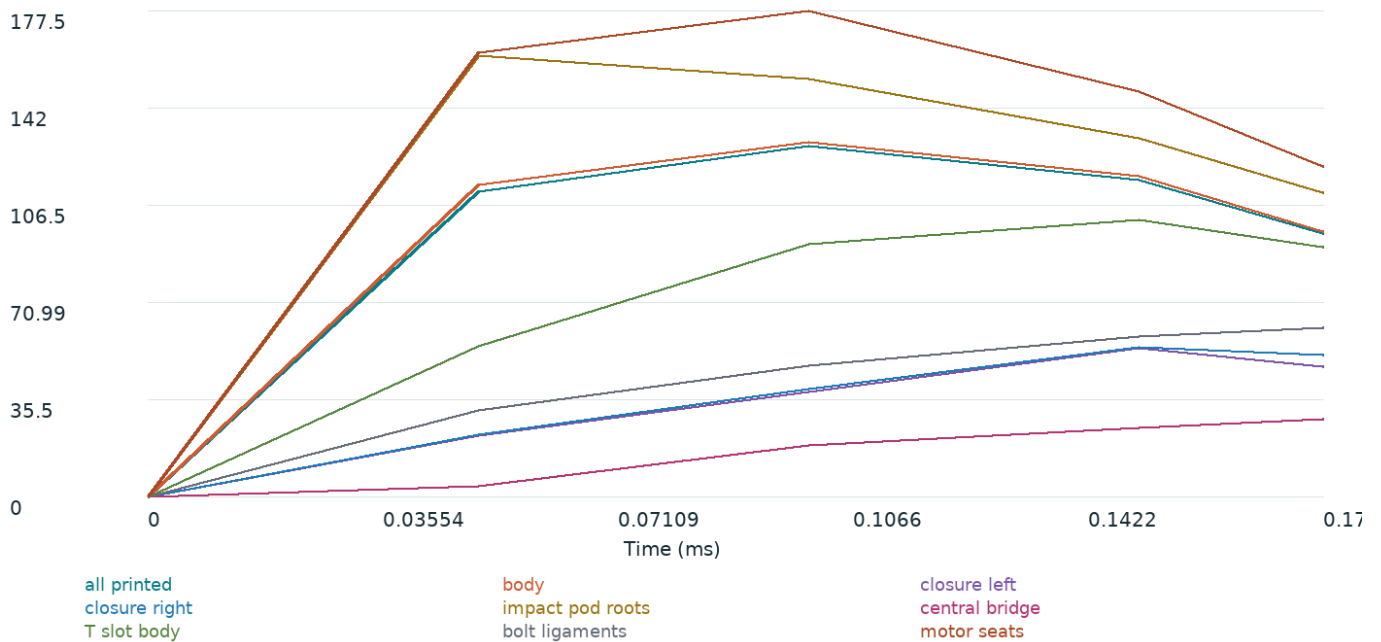
Compression (%)



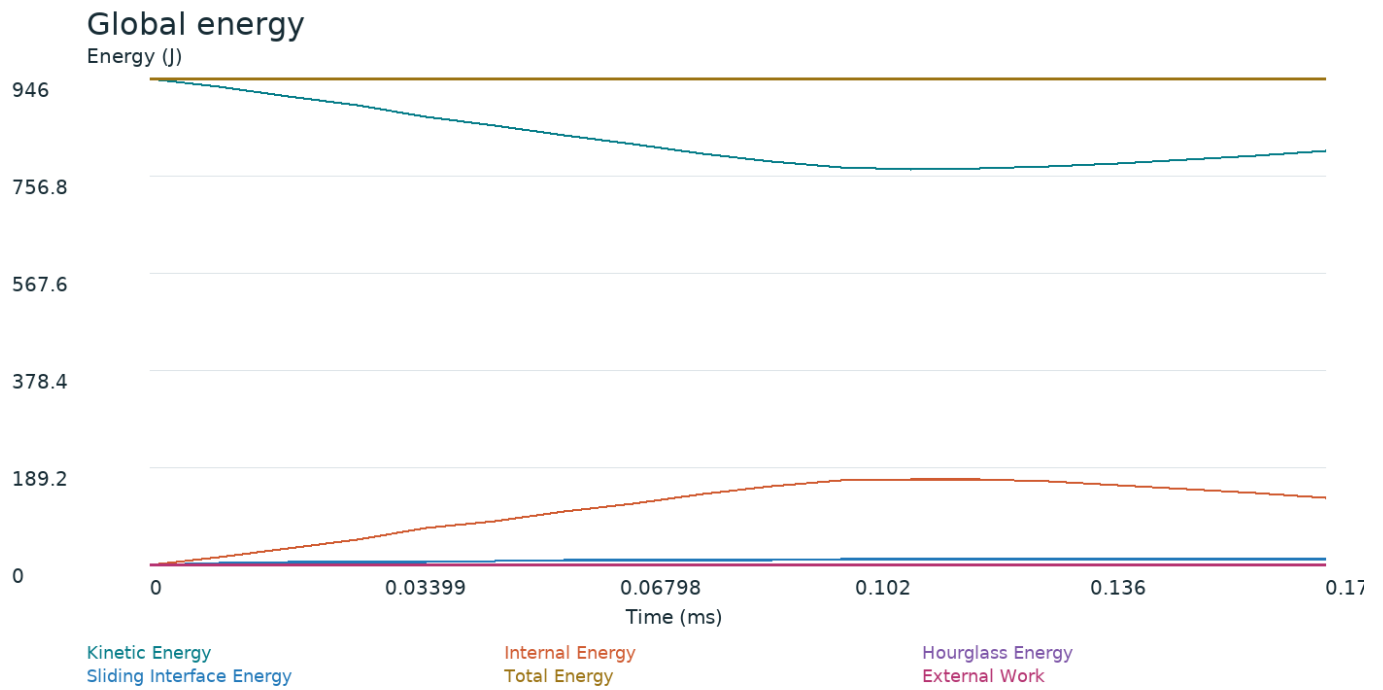
Foam compression. projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/time_histories.csv

Regional P99 von Mises stress

Stress (MPa)



Regional P99 von Mises stress.
projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/stress_histories.csv



Global energy. projects/local_closures_impact_20260927_01/cases/narrow-plug-5mm-v1/assessment/global_energy.csv

Results

Run 035 - impact evidence

Evidence status: stopped_early. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
assembly kind	local_closures
cad root	narrow_plug
printed kg	0.562764
hardware kg	0.0340022
motors kg	0.16
moving kg	0.756766
initial ke J	945.958
spreader each kg	0.135871

Solved mesh

Recorded mesh source: narrow_plug_5mm. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
Peak stopping force	98182.2	N
Maximum travel	7.34475	mm
Peak regional P99 (motor seats)	177.485	MPa
Last extracted time	0.177725	ms
Final forward velocity	28.4458	m/s
Energy balance error	0.00961988	%
Hourglass fraction	0	%

Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

Extracted numerical check	Evidence
normal termination	true
requested time reached	false
binary matches nodout	true
moving mass consistent	true
initial energy error below 0p5pct	true
energy balance below 5pct	true
hourglass below 5pct	true
nonfriction contact energy below 5pct	true
contact impulse error below 5pct	false
no axial spc on moving or guide	true
no detected deletion	true
positive foam jacobians	true
first stop captured	false
foam compression below 80pct	true
added mass below 0p1pct nominal	true
guide equations satisfied	false

Partial extraction: these histories do not establish a completed solve.

Provisional elastic demand only; mesh/timestep sensitivity, material calibration and fastener capacity unresolved.

Images, animations and source evidence

- contact_energy.csv
- contact_energy_totals.csv

- `contact_histories.csv`
- `global_energy.csv`
- `guide_equations.csv`
- `metrics.json`
- `stress_histories.csv`
- `time_histories.csv`

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.